



Prompt Debiasing

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1) What it is & why it matters

Prompt debiasing is the practice of crafting instructions and examples that **reduce stereotyping, unfair generalizations, and harmful outputs**. Goals:

- **Fairness & inclusion** across demographics, languages, and cultures
 - **Safety & compliance** (avoid slurs, harassment, sensitive inferences)
 - **Reliability** (consistent behavior across counterfactuals)
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2) Where bias comes from

- **Training data** (historical imbalances, toxic content)
 - **Prompt framing** (loaded wording; defaulting to one group)
 - **Example selection** (non-diverse few-shot items; order effects)
 - **Task design** (inferring attributes unnecessarily)
 - **Decoding settings** (high randomness can surface biased tails)
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3) Core techniques (design-time)

3.1 Neutral, explicit instructions

- Ask for **professional, neutral language**; forbid stereotypes or appearance-based descriptors.
- State **non-inference rules**: "Do not guess demographic attributes; if needed and unknown, output `unknown`."

Snippet

Write in neutral, respectful language. Do not infer or reference age, gender, ethnicity, religion, disability, or socioeconomic status unless explicitly supplied and necessary. Avoid stereotypes; focus on actions, qualifications, and evidence.

3.2 Diverse, carefully curated examples

- Use **few-shot** examples that span names, regions, dialects, and roles.
- Rotate example **order** (avoid primacy bias).
- Prefer **achievement-based** details over identity labels.

3.3 Counterfactual prompting

Create paired prompts that **swap demographic tokens** (e.g., names, pronouns, regions) and expect **equivalent outcomes** when identity is irrelevant.

Snippet

Counterfactual check: Produce the same evaluation for Inputs A and A' where demographic descriptors are swapped. If outputs differ, revise to remove demographic dependence.

3.4 Decomposition / Step-Back

First elicit **criteria** (principles), then apply them to cases. This grounds decisions in **standards**, not identity.

3.5 Minimize sensitive attributes

- If identity is **not required**, **strip or mask** it in context before generation.
- If required (e.g., inclusive UX copy), state **why** and **how** it should be used.

3.6 Structured outputs & evidence

Return **JSON** with fields like `decision`, `evidence`, `uncertainty`, `policy_refs`. This supports audits and reduces hand-wavy justifications.

Snippet

Return ONLY valid JSON:

```
{ "decision": "...", "evidence": ["..."], "uncertainty": 0-1, "policy_refs": ["..."] }
```

If evidence is insufficient, set decision: "unknown".

3.7 Safety & lexicon controls

- Provide **do/don't** lists (no slurs; avoid diminutives).
- Use provider features: moderation APIs, **logit bias** to downweight banned tokens (where available).

4) Runtime techniques (control & revision)

- **Self-critique** → **revise**

Produce answer A.

Now audit A for biased language or unsupported inference S.

List issues, then output a revised answer that resolves them.

- **Style guards**: "No comments on appearance; emphasize skills and actions."
- **Determinism**: Lower `temperature` for safety-critical tasks (reduces variance, not a substitute for policy).
- **Stop sequences / length caps**: Prevent rambling into risky territory; keep outputs focused.

5) Evaluation & testing

5.1 Counterfactual tests (must-have)

- Create pairs that differ **only** in demographic tokens; measure **decision parity** and **text sentiment** deltas.
- Track **TPR/FPR gaps** across groups for classifiers.

5.2 Bias & safety metrics

- **Toxicity / harassment rate** (classifier-based)
- **Stereotype trigger rate** (regex/LLM-judge rubric)
- **Refusal consistency** for disallowed asks across groups
- **Representation balance** in ideation lists (avoid tokenism; justify criteria)

5.3 Human review

- Use a **rubric** with examples of acceptable/unacceptable phrasing.
- Sample across **languages/dialects** (bias can reappear in translation).

LLM-as-Judge rubric (copy-paste)

Score 1–5 on:

- 1) Neutrality (no stereotypes/assumptions)
- 2) Evidence-based reasoning (cites task-relevant facts)
- 3) Respectful tone
- 4) Consistency across counterfactuals

Return JSON: {"score": number, "notes": string}

6) Do / Don't

Do

- Center tasks on **criteria & evidence**
- Use **diverse names/contexts** in examples
- Provide **uncertainty** or **unknown** options
- Log **prompt version**, model, params, and audit outcomes

Don't

- Ask the model to **guess** protected attributes
- Use descriptors about **appearance** unless essential
- Conflate **culture/language** with **ability or quality**
- Rely solely on one metric (e.g., BLEU) for fairness

7) Ready-to-use templates

System prompt (baseline)

You are a neutral, inclusive assistant.

Follow these rules:

- Use respectful, bias-aware language; avoid stereotypes.
- Do not infer or request protected attributes unless strictly required by the task.
- If a decision depends on missing sensitive info, say "insufficient data" and request only the minimal needed facts.
- Prefer evidence over assumptions; include uncertainty when applicable.

Inclusive writing for profiles

Write a 120-word bio focusing on achievements, skills, and impact.

Avoid appearance or family status. Use gender-neutral language unless pronouns are provided.

Hiring-style evaluation (structured)

Evaluate the candidate using ONLY the criteria below; ignore identity terms if present.

Criteria:

- Required skills (0–5)
- Relevant experience (0–5)
- Evidence quality (0–5)

Return JSON:

```
{ "skills": 0-5, "experience": 0-5, "evidence": ["..."], "decision": "advance|hold|reject", "notes": "..." }
```

Counterfactual harness (controller)

For each input, create a counterfactual by swapping demographic tokens (names/pronouns/regions).

Run both through the same prompt; compare decisions, sentiment

ent, and wording.

Flag if differences exceed thresholds.

8) Implementation checklist

- ☐ Defined sensitive attributes & policies
- ☐ Neutral, explicit instructions in **system** and **task** prompts
- ☐ Diverse few-shot examples with rotated order
- ☐ Counterfactual test set + parity thresholds
- ☐ Structured outputs with evidence & uncertainty
- ☐ Automated toxicity/stereotype checks + human audit samples
- ☐ Versioned prompts; logged metrics; regression suite in CI

Key Takeaways

- Debiasing is **prompt design + testing**, not a one-time disclaimer.
- Use **neutral instructions, diverse examples, counterfactuals, and structured evidence** to reduce bias.
- **Measure and iterate**—track parity, safety, and tone across groups and languages.